

Fall 2004

Problem 1 (Fa04). Show that there is a unique piecewise continuous function $y(x)$ on $\mathbb{R}$ satisfying

$$y(x) = \int_0^\infty e^{-2s} y(x-s) \, ds \quad \text{for } x > 0$$

$$y(x) = e^x \quad \text{for } x \leq 0$$

and find an explicit formula for $y(x)$ for $x > 0$.

Problem 2 (Fa04). For $c \in \mathbb{Q}$, define $R_c = \mathbb{Q}[x]/\langle x^3 - cx \rangle$. Let $a, b \in \mathbb{Q}$. Show that the rings R_a and R_b are isomorphic if and only if there exists a nonzero $r \in \mathbb{Q}$ such that $b = r^2 a$.

Problem 3 (Fa04). Let f and g be functions holomorphic on all of $\mathbb{C}$, except that g has an essential singularity at c . Prove that either f is constant, or the composition $f \circ g$ has an essential singularity at c .

Problem 4 (Fa04). Let A be an $n \times n$ matrix with complex entries. Prove that A is diagonalizable if and only if whenever f is a polynomial with complex coefficients such that $f(A)$ is nilpotent, we have $f(A) = 0$.

Problem 5 (Fa04). Let (a_m) be a sequence of real numbers satisfying $a_{n+m} \leq a_n + a_m$. Prove that

$$\lim_{n \rightarrow \infty} \frac{a_n}{n} = \inf_n \frac{a_n}{n}$$

Problem 6 (Fa04). Let n be a *squarefree* positive integer, i.e., $n = 1$ or n is prime or n is a product of distinct primes. Assume that for every product of primes $p q_1 \cdots q_r$ dividing n , with $r > 0$, we have $q_1 \cdots q_r \not\equiv 1 \pmod{p}$. Prove that every group G of order n is abelian.

Problem 7 (Fa04). Let $f: \mathbb{D} \rightarrow \mathbb{D}$ be a holomorphic function on the disk. Suppose that $f(-1/2) = 0$ and $f(0) = 1/2$. Prove that there is only one possible value for $f(1/2)$, and find it.

Problem 8 (Fa04). Let $\langle \cdot, \cdot \rangle$ be a positive definite Hermitian inner product on a complex vector space V . Suppose $T: V \rightarrow V$ is a $\mathbb{C}$ -linear map such that $\langle Tv, v \rangle = 0$ for all $v \in V$. Prove that $T = 0$.

Problem 9 (Fa04). Let $f: [0, 1] \rightarrow \mathbb{R}$ be a continuous function. Show that

$$\lim_{n \rightarrow \infty} \int_0^1 f(x) e^{inx^3} \, dx = 0$$

Problem 10 (Fa04). Let $m \leq n$.

1. Identify S_m with the subgroup of S_n of the elements that fix $m + 1, \dots, n$. Prove that $A_n \cap S_m = A_m$.
2. Is it true in general that if $f: S_m \rightarrow S_n$ is an injective homomorphism, then $A_n \cap f(S_m) = f(A_m)$? Give a proof or a counterexample.

Problem 11 (Sp09, Fa04). Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ be a continuous function of compact support. Show that

$$u(x) = \int_{\mathbb{R}^3} \frac{f(y)}{\|x - y\|} dy$$

is well defined and that

$$\lim_{\|x\| \rightarrow \infty} u(x)\|x\| = \int_{\mathbb{R}^3} f(y) dy$$

Problem 12 (Fa04). For which positive integers n does there exist an $n \times n$ matrix A with rational entries such that $A^3 + A + I = 0$?

Problem 13 (Fa04). Evaluate $I(w) = \int_0^\infty \frac{e^{iwt}}{\sqrt{t}} dt$ for every nonzero real number w .

Problem 14 (Fa04). What is the cardinality of the smallest field $\mathbb{F}$ of characteristic 7 such that the equation $x^{18} + x^{17} + \dots + x + 1 = 0$ has a root in $\mathbb{F}$?

Problem 15 (Fa04). Suppose that $f(z)$ is holomorphic on all of $\mathbb{C}$ except for a pole at $z = 0$. Prove that

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n f\left(\frac{1}{n} e^{2\pi i k/n}\right)$$

exists.

Problem 16 (Sp89, Fa97, Fa04). Let S be the subspace of $M_{n \times n}$ (the vector space of all real $n \times n$ matrices) generated by all matrices of the form $AB - BA$ with A and B in $M_{n \times n}$. Prove that $\dim S = n^2 - 1$.

Problem 17 (Fa04). A C^2 function $y(x)$ for $0 \leq x \leq 1$, a positive continuous function $a(x)$ for $0 \leq x \leq 1$, and a real number λ satisfy

$$y''(x) + \lambda a(x)y(x) = 0 \quad y(0) = 0 \quad y'(1) = 0$$

Suppose that $y(x)$ is not identically zero. Prove that $\lambda > 0$.

Problem 18 (Fa04). Prove that every group of order 30 has a cyclic subgroup of order 15.